

Canonical Quantization of Electromagnetic and Spinor Fields: Toward QED and Pseudo Quantum Electrodynamics

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Abstract

This paper aims to introduce the rigorous foundations of Quantum Electrodynamics (QED) and Quantum Field Theory (QFT) in general. It begins with the formalism of classical electrodynamics, as an example of a classical field theory. The formalism is then applied to a scalar field theory to show the procedure of quantizing a field. Based on these ideas, the paper proceeds to quantize the classical electromagnetic field. Next, the concept of the electron field is introduced, and its coupling to the quantized electromagnetic field is explored, which lead to the formulation of QED. The paper concludes with a discussion of a recent extension of QED applicable to condensed matter systems, known as **Pseudo Quantum Electrodynamics (PQED)**.

1 Introduction

The idea of fields that fill space and produce physical effects can be traced back to Michael Faraday in the mid-19th century. Through his inspired imagination, Faraday proposed that space is permeated by electric and magnetic fields, which mediate forces between charges. This conceptual framework was later formalized by James Clerk Maxwell in the 1860s, who unified electricity and magnetism into a single classical theory of electromagnetism. Maxwell's theory is regarded as the first successful example of a field theory.

Following the development of quantum mechanics in the early 20th century, physicists try hard to reconcile it with classical electrodynamics. This effort led to the formulation of Quantum Electrodynamics (QED), the quantum field theory of the electromagnetic interaction. There are two main approaches to QED. One approach begins with the classical field theory and proceeds by quantizing it using canonical quantization techniques, which is the method we will follow in this paper. The other approach is based on the path integral formalism, introduced by Richard Feynman in the 1940s, which provides an alternative for deriving quantum field theories.

Quantum Electrodynamics (QED) is the field theory that describes the interaction between charged particles and the electromagnetic field. While it was formulated in $(3+1)$ -dimensional spacetime, QED has found many applications beyond high-energy physics. In particular, it plays a role in modern condensed matter physics, especially in the study of two-dimensional materials such as graphene. Consequently, the dimensional reduction of QED from $(3+1)$ to $(2+1)$ dimensions has emerged as an essential area of research, allowing for more accurate modeling.

This paper is organized as follows. In Section 2, we formulate classical electrodynamics using the Lagrangian formalism. In Section 3, we apply canonical quantization to a scalar field and then generalize the procedure to the electromagnetic field. In Section 4, we introduce the spinor field and proceed to quantize it. In Section 5, we consider the interaction between the spinor and electromagnetic fields, leading to

the formulation of Quantum Electrodynamics (QED). Finally, in Section 6, we discuss Pseudo Quantum Electrodynamics (PQED) as a dimensional reduction of QED applicable to two-dimensional systems.

2 Electromagnetism as a Classical Field Theory

A possible description of the electromagnetism is as a *massless vector field theory*. That is, for each point in space-time $x_\mu = (x_1, x_2, x_3, x_4)^T$ there is a corresponding 4-vector $A_\mu = (\frac{V}{c}, \mathbf{A})$ where c is the speed of light and $V, \mathbf{A}$ are the scalar and vector potential, respectively.

A full description of the electric $\mathbf{E}$ and the magnetic field $\mathbf{B}$ can be obtained if we get the 4-vector field A_μ . Indeed, we have

$$E_i = -\partial_i A_0 - \partial_0 A_i \quad (1)$$

$$B_i = \epsilon_{ijk} \partial_j A_k \quad (2)$$

where $i, j, k = 1, 2, 3$ and ϵ_{ilk} is the Levi-Civita symbol.

We define the electromagnetic field tensor (or the field strength tensor) as

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

Since it is fully described by the 4-vector field A_μ , we can get the electric and magnetic field in terms of the field strength tensor as follows :

$$E_i = cF_{0i} \quad (3)$$

$$B_i = -\frac{1}{2}\epsilon_{ijk}F^{jk} \quad (4)$$

The Lagrangian density of the electromagnetic field in the absence of any sources is given [1, 4] by

$$\mathcal{L} = \frac{-1}{4}F_{\mu\nu}F^{\mu\nu} \quad (5)$$

thus, by minimizing the corresponding action we get the following equation of motion[1, 3, 4]:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) = -\partial_\mu F^{\mu\nu} = 0 \quad (6)$$

which from we can "derive" the famous *Maxwell's equations*.

A very important property of the Lagrangian density given in (5) is that it is invariant under the *gauge symmetry* which means

$$F_{\mu\nu} \rightarrow \partial_\mu (A_\nu + \partial_\nu \lambda) - \partial_\nu (A_\mu + \partial_\mu \lambda) = F_{\mu\nu} \quad (7)$$

where λ is any well-behaved function. Thus, the electromagnetic field is represented by a **massless gauge field theory**.

3 The Quantization of the Electromagnetic Field

In this paper, we apply the *canonical quantization* formalism to the quantization of arbitrary fields. In this section, we will first apply the method to a scalar field and then generalize it to the electromagnetic field.

3.1 Quantization Scalar Field

To illustrate the process of quantization, it is instructive to begin with the simple example of a scalar field. A scalar field $\phi(x_\mu)$ can be viewed as a real-valued function on spacetime, where each point x corresponds to a harmonic oscillator. Thus, the field can be visualized as a continuous collection of coupled harmonic oscillators distributed throughout space. The dynamics of the field are governed by the Lagrangian density[4]:

$$\mathcal{L}_{\text{scalar}} = \frac{1}{2} (\partial^\mu \phi \partial_\mu \phi - m^2 \phi^2), \quad (8)$$

where m is an intrinsic parameter associated with the field.

Quantization proceeds by:

- promoting the classical dynamical variables to operators
- imposing suitable commutation relations

analogous to the quantization of a single harmonic oscillator. This framework naturally leads to the introduction of *creation and annihilation* operators, which can be interpreted as adding or removing a single photon from the field.

First, let us define the analog of momentum, known as the conjugate momentum, by

$$\pi(x_\mu) = \frac{\delta \mathcal{L}}{\delta \dot{\phi}(x_\mu)} = \dot{\phi}(x_\mu),$$

where $\dot{\phi}(x^\mu)$ denotes the derivative of the field with respect to time.

We then promote the canonical variables $\phi(x_\mu)$ and $\pi(x_\mu)$ to operators,

$$\phi(x_\mu) \rightarrow \hat{\phi}(x_\mu), \quad \pi(x_\mu) \rightarrow \hat{\pi}(x_\mu) \quad (9)$$

Next, we impose commutation relation on them

$$[\phi(x_\mu), \pi(y_\mu)] = i\delta^3(\mathbf{x} - \mathbf{y}), \quad (10)$$

$$[\phi(x_\mu), \phi(y_\mu)] = 0, \quad [\pi(x_\mu), \pi(y_\mu)] = 0 \quad (11)$$

where $\delta^3(\mathbf{x} - \mathbf{y})$ is the three-dimensional Dirac delta function.

By utilizing these commutation relations, we can expand the field in terms of creation and annihilation operators, revealing the quantized nature of the field.

The field operator $\hat{\phi}(x_\mu)$ can be written as [1, 3]:

$$\hat{\phi}(x_\mu) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} (\hat{a}(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x} - iE_{\mathbf{p}}t} + \hat{a}^\dagger(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{x} + iE_{\mathbf{p}}t}), \quad (12)$$

where $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$ is the relativistic energy.

The operators $\hat{a}(\mathbf{p})$ and $\hat{a}^\dagger(\mathbf{p})$ satisfy the commutation relations

$$[\hat{a}(\mathbf{p}), \hat{a}^\dagger(\mathbf{p}')] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{p}'), \quad (13)$$

$$[\hat{a}(\mathbf{p}), \hat{a}(\mathbf{p}')] = [\hat{a}^\dagger(\mathbf{p}), \hat{a}^\dagger(\mathbf{p}')] = 0. \quad (14)$$

Hence, we have fully quantized the scalar field $\phi(x_\mu)$. The field operators are expressed in terms of creation and annihilation operators, which satisfy canonical commutation relations. These operators encode the quantum excitations of the field, corresponding to particles with definite momenta and energies.

3.2 The Quantization of the Electromagnetic Field

The electromagnetic field is a gauge field that is fully described by the four-vector $A_\mu(x_\mu)$, which is a function of spacetime. To quantize the electromagnetic field, we follow an approach similar to that used for the scalar field. First, we need to identify the conjugate momentum. Using the Lagrangian in equation (5), we find

$$\pi^i(x_\mu) = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_i)} = -E_i(x_\mu),$$

However, we encounter an important subtlety: the time component A_0 does not possess a conjugate momentum. This reflects the fact that the electromagnetic field has gauge redundancy, and not all components of A_μ correspond to independent physical degrees of freedom [4].

To proceed with quantization, we must fix a gauge to eliminate this redundancy. A common choice is the Coulomb gauge, defined by the condition

$$\nabla \cdot \mathbf{A} = 0 \quad (15)$$

After fixing the gauge, we promote the remaining dynamical variables to operators and impose canonical commutation relations on the physical degrees of freedom.

In the Coulomb gauge, the dynamical degrees of freedom are the two transverse components of the vector potential $\mathbf{A}(x_\mu)$. We impose the following commutation relations:

$$[A_i(x_\mu), \pi_j(y_\mu)] = i\delta_{Tij}(\mathbf{x} - \mathbf{y}),$$

$$[A_i(x_\mu), A_j(y_\mu)] = [\pi_i(x_\mu), \pi_j(y_\mu)] = 0,$$

where $\delta_{Tij}(\mathbf{x} - \mathbf{y})$ is the transverse delta function, projecting onto the physical (transverse) degrees of freedom.

The quantized vector potential $\mathbf{A}(x_\mu)$ can be expanded in terms of creation and annihilation operators as [1, 3]:

$$\mathbf{A}(x_\mu) = \sum_{\lambda=1}^2 \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(\hat{a}_{\mathbf{p},\lambda} \boldsymbol{\epsilon}_\lambda(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x} - iE_{\mathbf{p}}t} + \hat{a}_{\mathbf{p},\lambda}^\dagger \boldsymbol{\epsilon}_\lambda^*(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x} + iE_{\mathbf{p}}t} \right), \quad (16)$$

where:

- $E_{\mathbf{p}} = |\mathbf{p}|$ is the energy of a photon with momentum $\mathbf{p}$,

- $\epsilon_\lambda(\mathbf{p})$ are transverse polarization vectors satisfying

$$\mathbf{p} \cdot \epsilon_\lambda(\mathbf{p}) = 0,$$

- the operators $\hat{a}_{\mathbf{p},\lambda}$ and $\hat{a}_{\mathbf{p},\lambda}^\dagger$ obey the commutation relations

$$[\hat{a}_{\mathbf{p},\lambda}, \hat{a}_{\mathbf{p}',\lambda'}^\dagger] = (2\pi)^3 \delta_{\lambda\lambda'} \delta^3(\mathbf{p} - \mathbf{p}'),$$

$$[\hat{a}_{\mathbf{p},\lambda}, \hat{a}_{\mathbf{p}',\lambda'}] = [\hat{a}_{\mathbf{p},\lambda}^\dagger, \hat{a}_{\mathbf{p}',\lambda'}^\dagger] = 0.$$

This means that the vector potential is described in terms of the creation and annihilation of discrete numbers of photons, represented by the corresponding creation and annihilation operators.

4 Quantization of Spinor Fields

As our final goal is to develop a quantum theory that describes the interaction between the electron and the electromagnetic field, we need a field that represents the electron. This is achieved using the **spinor field**. The spinor field describes particles with spin- $\frac{1}{2}$, such as electrons and quarks.

The field is represented by a four-component spinor $\psi(x_\mu)$, which is a function of spacetime.

The dynamics of the free spinor field are governed by the Dirac Lagrangian[1, 3, 4]:

$$\mathcal{L}_{\text{spinor}} = \bar{\psi}(x_\mu)(i\gamma^\mu \partial_\mu - m)\psi(x_\mu), \quad (17)$$

where $\bar{\psi}(x_\mu) = \psi^\dagger(x_\mu)\gamma^0$ is the Dirac adjoint, m is the mass of the particle, and γ^μ are the gamma matrices

To quantize the spinor field, we first identify the *conjugate momentum* associated with $\psi(x_\mu)$:

$$\pi(x_\mu) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi(x_\mu))} = i\psi^\dagger(x_\mu).$$

Now, we promote the fields $\psi(x_\mu)$ and $\pi(x_\mu)$ to operators and impose canonical *anticommutation relations* (because ψ is a fermionic field):

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta^\dagger(\mathbf{y}, t)\} = \delta_{\alpha\beta} \delta^3(\mathbf{x} - \mathbf{y}),$$

$$\{\psi_\alpha(\mathbf{x}, t), \psi_\beta(\mathbf{y}, t)\} = \{\psi_\alpha^\dagger(\mathbf{x}, t), \psi_\beta^\dagger(\mathbf{y}, t)\} = 0,$$

where α, β are spinor indices.

The spinor field operator $\psi(x_\mu)$ can be expanded in terms of plane-wave solutions as [1]:

$$\psi(x_\mu) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \sum_{s=1}^2 (b_s(\mathbf{p})u_s(\mathbf{p})e^{-ip \cdot x} + d_s^\dagger(\mathbf{p})v_s(\mathbf{p})e^{ip \cdot x}),$$

where:

- $u_s(\mathbf{p})$ and $v_s(\mathbf{p})$ are positive- and negative-energy spinor solutions of the Dirac equation,
- $b_s(\mathbf{p})$ annihilates a particle (fermion) with momentum $\mathbf{p}$ and spin s ,

- $d_s^\dagger(\mathbf{p})$ creates an antiparticle (antifermion) with momentum $\mathbf{p}$ and spin s ,
- $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$.

The operators satisfy the anticommutation relations:

$$\{b_s(\mathbf{p}), b_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta_{ss'} \delta^3(\mathbf{p} - \mathbf{p}'),$$

$$\{d_s(\mathbf{p}), d_{s'}^\dagger(\mathbf{p}')\} = (2\pi)^3 \delta_{ss'} \delta^3(\mathbf{p} - \mathbf{p}'),$$

with all other anticommutators vanishing.

The Hamiltonian operator can be written in terms of the creation and annihilation operators as[4]:

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 (b_s^\dagger(\mathbf{p}) b_s(\mathbf{p}) + d_s^\dagger(\mathbf{p}) d_s(\mathbf{p})).$$

Despite the mathematical formalism, the physical picture is that an electron can be thought of as an excitation of the corresponding spinor field. The creation operators provide a precise representation of this excitation. In this framework, all electrons in the universe are understood as quantized excitations of the underlying spinor field.

5 Quantum Electrodynamics (QED)

Having now described the electron in terms of a *quantized* spinor field and the photon in terms of the *quantized* electromagnetic field, the next step is to examine the interaction between them. This coupling forms the basis of what is known as **quantum electrodynamics (QED)**.

One might reasonably guess that the Lagrangian of quantum electrodynamics (QED) is simply the sum of the Lagrangians for the spinor field and the electromagnetic field, along with an interaction term. That is:

$$\mathcal{L}_{\text{QED}} = \mathcal{L}_{\text{Maxwell}} + \mathcal{L}_{\text{spinor}} + \text{interaction}.$$

Indeed, it turns out that the interaction term is given by[1]:

$$-e\bar{\psi}\gamma^\mu\psi A_\mu,$$

where e quantifies the strength of the interaction between the two fields. In the context of QED, e corresponds to the electric charge of the electron.

Thus, we can write the QED Lagrangian in a more compact form:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(iD_\mu - m)\psi, \quad (18)$$

where the covariant derivative is defined as

$$D_\mu = \partial_\mu + ieA_\mu,$$

commonly referred to as the *covariant derivative*.

Note that under a local $U(1)$ gauge transformation,

$$\psi \rightarrow e^{ie\lambda(x)}\psi, \quad A^\mu \rightarrow A^\mu - \partial^\mu \lambda,$$

for a well-behaved function $\lambda(x)$, the Lagrangian remains invariant. This gauge invariance under local $U(1)$ transformations is a defining feature of QED, identifying it as a $U(1)$ gauge theory.

Quantum Electrodynamics (QED) is relevant to any field of physics that deals with the interaction between electrons and photons. Despite its complexity, it is considered the most accurate theoretical description of such interactions. In many cases, the theory can be significantly simplified by employing suitable approximations or by applying dimensional reduction techniques.

6 Pseudo-Quantum Electrodynamics(PQED)

In some physical applications, we encounter situations where the interaction between electrons is confined to planar materials, such as graphene. One approach is to consider a (2+1)-dimensional QED theory to describe such a system [2]. However, the problem with this approach is that it yields in the static limit a potential between the electrons, $V(r)$, which behaves as $\ln(r)$ [8]. This clearly indicates that the (2+1) QED model is not valid for describing the interaction in such materials as it does not obey the coulomb potential which goes as $\frac{1}{r}$.

This discrepancy arises from incorrectly assuming that the propagation of the electromagnetic field is confined strictly to the plane. However, this is not the case. Although the kinematics of the electrons are restricted to a two-dimensional material, the electromagnetic field propagates throughout the full three-dimensional space[7]. A field theory that treats the spinor field in $(2+1)$ -dimensions while allowing the electromagnetic field to remain in $(3+1)$ -dimensions is known as **Pseudo-Quantum Electrodynamics (PQED)**.

A dimensionality reduction technique can be used [5, 7] to derive the Lagrangian density that describes the model of Pseudo-Quantum Electrodynamics (PQED):

$$\mathcal{L}_{\text{PQED}} = \frac{1}{2} \frac{F_{\mu\nu} F^{\mu\nu}}{(-\square)^{1/2}} + \mathcal{L}_{\text{spinor}} + j^\mu A_\mu - \frac{\zeta}{2} \frac{\partial^\mu \partial^\nu}{(-\square)^{1/2}} A_\mu, \quad (19)$$

where $\square = \partial^\mu \partial_\mu$ is the d'Alembert operator, and ζ is a gauge-fixing parameter.

The pseudo-differential operator $(-\square)^{-1/2}$ actually appeared prior to the formal development of PQED. Notably, it was introduced in the 1990s in the context of certain superconductivity models[9]. In the case of PQED, the operator $(-\square)^{-1/2}$ arises as a consequence of the fact that the electromagnetic field propagates in three spatial dimensions, while the spinor fields are confined to a two-dimensional surface [5, 6].

Due to the presence of the pseudo-differential operator $(-\square)^{-1/2}$, the Lagrangian in equation (19) is *nonlocal* and Consequently, PQED is a nonlocal field theory. Despite this nonlocality, the resulting theory correctly predicts a $1/r$ Coulomb potential in the plane, which is consistent with experimental observations in two-dimensional materials such as graphene [6]. In particular, it has been shown that the static potential derived from PQED is [10]:

$$V(r) = \frac{e^2}{4\pi(1+\theta^2)} \frac{1}{r} \quad (20)$$

where θ is a parameter related to the properties of the material.

Despite the nonlocality of equation (19), PQED remains consistent with quantum mechanics. In particular, it has been demonstrated that the S -matrix of the theory is unitary[11], which implies that the sum of probabilities is conserved and equals one—fully aligning with the probabilistic interpretation of quantum mechanics. In addition, a physical theory must also respect *causality*. Indeed, it has been shown in [12] that PQED respects causality.

Recent developments in the study of topological phases of matter have showed that the Pseudo Quantum Electrodynamics (PQED) theory provides an effective theoretical description of the Quantum Valley Hall effect observed in graphene superlattices[13] and monolayers of SiC [14]. These findings highlight the relevance of PQED not only in planar quantum field theories but also in the broader context of topologically nontrivial quantum systems.

7 Conclusion

To conclude, we began with classical electrodynamics and applied the formalism of canonical quantization to obtain a gauge quantum field theory. We then introduced a spinor field and coupled it to the gauge field, resulting in Quantum Electrodynamics (QED), a $U(1)$ gauge theory. By performing a dimensional reduction—restricting the spinor fields to a two-dimensional plane while allowing the electromagnetic field to propagate in three dimensions—we arrived at Pseudo Quantum Electrodynamics (PQED). We have seen that PQED correctly reproduces the Coulomb potential in the plane and remains consistent with both quantum mechanics and special relativity. Finally, we discussed recent research developments related to PQED.

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